Biomedical Engineering Laboratory I (Fall 2025)

Lab 2 Biomechanical Testing[1]

OBJECTIVES

1. Learn basic mechanics and mechanical testing concepts including the fundamentals of tensile, compressive, and three-point bending testing.
2. Be familiar with techniques that used in load and deformation measurements.
3. Understand the difference between structural and material properties.
4. Empirically determine the elastic and yield properties of various materials.
5. Learn the basic structural and functional components of bones.

BACKGROUND

Knowledge of the mechanical properties of engineering materials and biological tissues is important for the design of functional artificial tissues, determination of normal physiological functions, and assessment of changes in tissues that might arise due to disease. Mechanical properties can be assessed in a laboratory using a variety of techniques. In this module you will perform tensile testing and three-point bending testing of metal, plastic specimen, and hard biological tissues (bone).

Mechanical Testing – Tensile Testing

Amongst the most common methods of assessing material properties is tensile testing. During tensile testing, a specimen of a standard shape (usually a dog bone shape) and dimensions is subjected to an axial load. The specimen is gripped at its two ends and is pulled to elongate at a specific deformation rate to its breakpoint. During testing, the axial load (P) applied to the specimen (in Newtons, N) and the deformation in the specimen are monitored. The load is used to determine the stress, σ (N/m^2) in the specimen:

Figure 1. Material undergoing tensile testing

[1] Adapted from Prof. Aaron M. Kyle, Columbia University.

$$\sigma = \frac{P}{A} \quad (1)$$

where A is the cross-sectional area of the specimen. Material deformation is also monitored during testing, the strain in the material (ε) can be determined:

$$\varepsilon = \frac{\Delta L}{L_0} \quad (2)$$

where ΔL is the change in specimen length with respect to original length (original gauge length), L_0. The strain is a dimensionless parameter that quantifies a material's deformation in response to an applied load.

The deformation and load in a material undergoing tensile testing are used to form an analytical curve called a **stress-strain plot**. In these plots, the monitored stress in the material is plotted as a function of the induced strain (a representative stress-strain plot shown in Figure 2.) There is a great deal of material-specific information that is derived from the stress-strain plot; such testing is often the basis for material characterization. For the purposes of biomechanical testing in this lab, we are chiefly concerned about the following.

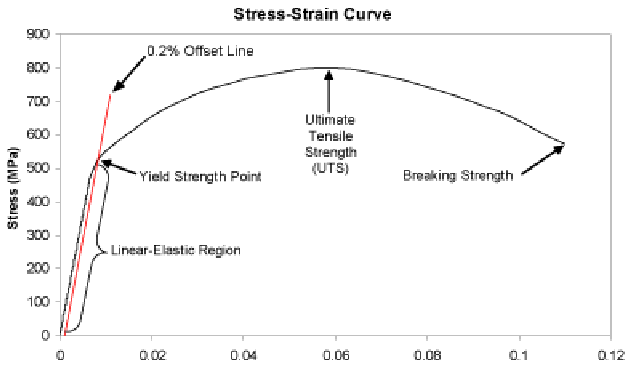

Figure 2. Representative stress-strain plot for material undergoing tensile testing

- *Young's (Elastic) Modulus*

The linear region of the stress-strain plot is called the **elastic region**. When the material is deformed in this region, it will return to its original shape when the force is removed. The slope of the curve in this region corresponds to a material property known as the **Young's modulus**, E, also known as the elastic modulus. Therefore, the elastic modulus can be determined from the linear phase of the material's stress-strain response:

$$\sigma = E\varepsilon \quad (3)$$

The Young's modulus can be used to describe material's response to loading. For example, hard materials, like metals or ceramics that are used in orthopedic implants, have a large Young's modulus due to their ability to withstand large stress, while exhibiting small deformations (strains). Conversely, soft materials, like blood vessels or other soft tissues, will exhibit low stress in response to applied deformation, i.e., they will readily deform without inducing much stress.

- *Plastic Region*

A material can only deform elastically within a certain range of stresses. With increasing stress, the molecular structure of the material changes, resulting in plastic deformation. In this region, the stress- strain relationship is no longer linear and the material undergoes permanent deformation. The point at which deformation changes from elastic to plastic is denoted by the **yield strength** (σ_y). The yield strength is often reported as a material property, indicating the relative ductility of a material. The stress at which yield occurs is dependent on both the rate of deformation (strain rate) and, more significantly, the temperature at which the deformation occurs. Generally, the yield strength increases with strain rate and decreases with temperature. Yield strength may be determined as the point where the stress-strain curve changes from linear to non-linear behavior, which is the approach for polymers and soft biological materials. For metals, a method called the 0.2% offset method is often used to estimate σ_y. To determine this value from a stress-strain plot, a line is drawn from 0.2% strain point whose slope is equal to E. The intersection of this line and the material's stress-strain curve is identified as the 0.2% offset σ_y. However, the method for determining yield strength varies from universal standards provided by different professional societies.

- *Ultimate Tensile and Fracture Strengths*

The maximum stress that a material can withstand is denoted as the **ultimate tensile strength (UTS)**. In brittle materials, the UTS will be at the end of the linear-elastic portion of the stress-strain curve or close to the elastic limit. In ductile materials, the UTS will be well outside of the elastic portion into the plastic portion of the stress-strain curve. The **fracture strength** is the point of failure for a material, i.e., the material breaks and energy of deformation is rapidly released. Often in brittle materials, the fracture strength and UTS will be the same.

While tensile testing is commonly used for evaluating materials, particularly metals and inorganics, it is not appropriate for testing biological samples. In particular, bones, due to their irregular shapes and different mechanical behaviors in tension, compression, and varying orientations (anisotropy) are not well-served by tensile testing. In order to conducting rigorous mechanical characterization of bone and similar tissues, a different modality is used: Three Point Bending.

Mechanical Testing: Three-Point Bending

The three-point bending testing configuration is another common testing technique used for determining material properties of both inorganic and biological samples. Three-point bending is particularly advantageous for characterization of biological structures, e.g., explanted tissues as it requires minimal specimen preparation and can be easily performed. In general, a long specimen is placed under an upper loading post while supported by two lower supporting posts at the ends of the specimen (see Figure 3). The upper loading post of the three-point fixture is connected to the actuator of the device, which has transducers for displacement measurement and load measurement respectively.

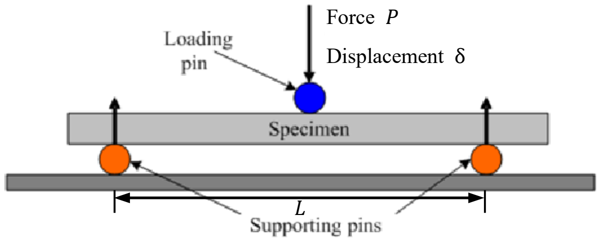

Figure 3. Schematic diagram of 3-point bending test

Accordingly, the deflection of the specimen (δ) and the load (Force) applied to the specimen during testing can be determined throughout the load testing. Note that because of the orientation, the deflection is not directly applied to determine strain as in uniaxial testing; the actual determination of strain will be explained later in this section.

The shear force and moment distributions in the specimen during three-point testing can be analyzed using a free body diagram as shown in Figure 4. The maximum moment (M_{max}) occurs at the mid-span of the specimen ($x = L/2$) with a magnitude of:

$$M_{max} = \frac{PL}{4} \qquad (4)$$

At any cross-section, the specimen is subjected to bending. In beam theory, the bending stress, or the

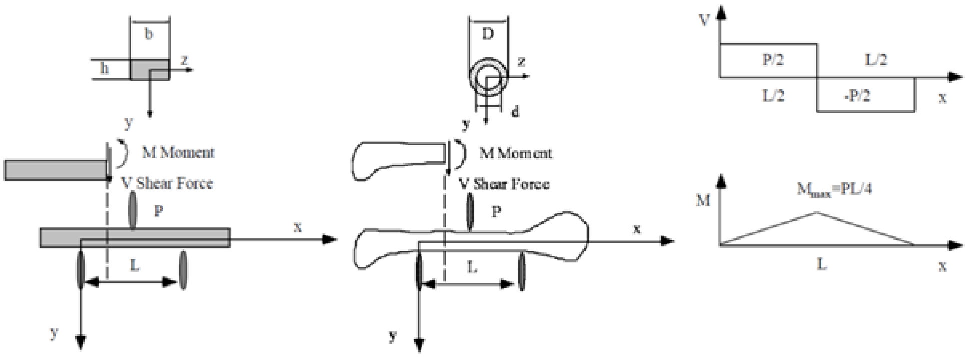

Figure 4. Shear forces and moments associated 3-Point Bending Test of a specimen with a rectangular and a cylindrical hollow cross section

stress in the axial direction (x), has a linear distribution along the y-axis with compressive stress in the top half of the cross-section and tensile stress in the bottom half. The bending stress (σ) varies along the cross section of the specimen and is related to the bending moment by:

$$\sigma = \frac{M}{I} y \tag{5}$$

where I is the moment of inertia of the cross-section. The moment of inertia is calculated based on the geometry of the specimen. I is calculated for a solid rectangular cross-section using:

$$I = \frac{bh^3}{12} \tag{6}$$

where b and h are the width and height of the specimen, respectively. For a hollow specimen with a cylindrical cross section, the moment of inertia (I_{cyl}) is calculated by:

$$I_{cyl} = \frac{\pi}{64}(D^4 - d^4) \tag{7}$$

where D and d are the outer and inner diameters of the specimen, respectively. You will use Eqn. 7 to determine I for a segment of cortical bone. The maximum stress (σ_{max}) occurs at the outer surface at the middle of the specimen ($y = h/2$ for a rectangular specimen), where $M = M_{max}$. For a specimen with a rectangular cross-section.:

$$\sigma_{max} = \frac{M_{max} h}{2I} \tag{8}$$

For a cylindrical structure:

$$\sigma_{max-cyl} = \frac{MD}{2I} \tag{9}$$

The maximum strain (ε_{max}) also occurs at the outer surface:

$$\varepsilon_{max} = \frac{\sigma_{max}}{E} = \frac{PLh}{8EI}, or, \frac{PLD}{8EI_{cyl}} \tag{10}$$

During a three-point bending experiment, you will derive a load-deflection curve as shown in Figure 5. The load is proportionate to the deflection in the specimen:

$$P = \frac{48EI}{L^3}\delta \tag{11}$$

By fitting a line to the linear portion of the load-deflection curve, the slope (**stiffness, S**) can be determined. The stiffness is a **structural property**, which means that its value is dependent on the intrinsic properties of the material as well as its geometry. Re-arranging Eqn. 11 and knowing the cross-sectional geometry and the span of the specimen, the Young's modulus can be determined:

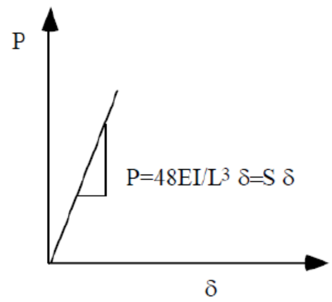

Figure 5. Representative load-deflection curve.

$$E = \frac{PL^3}{48\delta I} \tag{12}$$

Note that the Young's modulus is a **material property**; it is an intrinsic characteristic of the material that quantifies the material's elasticity regardless of geometry. In this procedure, you will empirically determine the load-deflection curves for metal, polymer, and biological tissue (cortical bone) specimens. From these curves and using analysis similar to that described above, you will determine the structural and material properties of the various specimens.

MATERIALS AND METHODS

- Metal rods for tensile test and metal bars/beams for bending test: cast iron (HT150), low carbon steel (Q235)
- Polymer bars for tensile test: PS (polystyrene), PE (polyethylene)
- Raw and cooked chicken drumsticks
- Calipers
- Universal material testing machine
- Two accessories for universal material testing machine: Extensometer and Long-travel extensometer

LABORATORY PROCEDURES

You should wear protective goggles during testing. Stand a safe distance away from the equipment and do not attempt to adjust or move a specimen while testing. Keep your hands, arms and elbows away from the test area.

The load-displacement(deformation) data will be stored in the computer and should be copied immediately after your testing for post-lab analysis. Data processing can be performed using MATLAB, Excel, or SPSS and so on.

Procedure 1: Tensile tests – metal

In this procedure, two kinds of metal rod specimens, cast iron (HT150) and low carbon steel (Q235), of circular cross section will be subjected to tensile test; two strain rates (by two crosshead speed) will be applied to them. A range of universal standards provided by professional societies such as American Society of Testing and Materials (ASTM), British standard, DIN standard and Chinese GB Standard are selected based on preferential uses. Each standard may contain a variety of test standards suitable for different materials, dimensions and fabrication history. In this procedure, the specimens are prepared based on Chinese GB Standard GB/T 228.1 [1].

1. For each member of your group, fetch a piece of cast iron rod specimen for tensile test from the container. Use a caliper to measure the diameter d on a cross section in two perpendicular directions, and then perform this measurement three times along the length of the specimen. The designed gauge length will be given in the class. *Record the average dimensions and use these values as the geometric data for the specimen*.
2. Mount the specimen on a tensile test fixture, then fix the extensometer onto the specimen just as shown by TA.
3. Check the tensile testing protocol to ensure it is the same as the instructor-provided specifications. At the first time, the test will be conducted in the specified low crosshead speed (producing a low strain rate). *In your lab report, provide the crosshead speed and the resultant strain rate.*
4. Run the testing protocol until test terminates, usually after the specimen fractures.
5. Rename and save the testing data file right after the testing.
6. Remove the extensometer from the specimen and place it on the platform of the testing machine, then remove the fractured specimen from the test fixture.
7. Repeat steps 1-6 on one more cast iron rod, change the crosshead speed to the specified higher one (producing a higher strain rate).
8. Repeat steps 1-7 on two low carbon steel specimens.
9. *Describe how the specimen's shape changes during the test and what that implies?*
10. *For each specimen, create a stress-strain plot using the load-deformation data; determine the Young's modulus, yield strength (upper and lower if applicable) and ultimate tensile strength of the specimen from the resultant plot. For each testing condition, you will have 3 or 4 plots in your group; Provide and*

comment on one representative stress-strain plot for each condition.

11. *The resultant Young's moduli, yield strengths and ultimate tensile strengths from all the groups will be pooled for statistical analysis. (Statistically) Compare these results from cast iron to low carbon steel, low strain rate to high strain rate. Using appropriate statistical testing to demonstrate whether there are differences in the measured properties (1) between two kinds of materials and (2) when the strain rate is changed.*
12. *Are yield strength and ultimate tensile strength structural or material properties of the sample? State your reason in lab report.*
13. *Search the literature and material databases for the published properties mentioned above of the materials used in this experiment. Quantitatively compare your measured values to the published values (be sure to cite your sources).*

Procedure 2: Tensile tests – polymer

In this procedure, PS (polystyrene) and PE (polyethylene) tensile bars with rectangular cross sections will be subjected to tensile test. As you can see, the fixtures for bar tensile test are different from those for rod. In this procedure, the specimens are prepared based on Chinese GB Standard GB/T 1040.2 [2]. For each member in your group, the following steps should be performed on one PS and one PE tensile bar.

1. Fetch a piece of tensile bar specimen from the container. Use a caliper to measure the width, b, and the height, h, of the specimen. Perform this measurement three times along the length of the specimen. According to the specimen's design, the original gauge length is 50.0 mm. *Record the average dimensions and use these values as the geometric data for the specimen.*
2. The other steps will be similar to those in Procedure 1, with the exception of keeping the crosshead speed constant at 30 mm/min and using a long travel extensometer to measure the exact deformation of the plastic. The plastic specimen may not fracture when the test was terminated, because the elongation of some plastic go beyond the limit of the testing machine.

Procedure 3: 3-Point Bending Test – metal

In this procedure, cast iron and low carbon steel bar/beam specimens of rectangular cross section will be subjected to 3-point bending test.

1. Measure the width b and the height h of the specimen. Perform this measurement three times

along the length of the specimen. *Record the average dimensions and use these values as the geometric data for the specimen*.

2. Measure the span L, which is the distance between two supporting posts on the testing unit.
3. Place the specimen on the 3-point bending test fixtures.
4. Using the testing machine software, lower the upper loading post until it almost touches (but does not touch) the specimen, leaving a distance of 1-2 mm. Check the testing protocol to ensure it is the same to the instructor-provided specifications. *In your lab report, provide the crosshead speed.*
5. Run the testing protocol until test terminates.
6. Rename and save the testing data file immediately after the test.
7. *From the load-deflection data, use beam theory to determine the maximum stress in the specimen varying with deflection(eq.8).*
8. *Determine the Young's modulus E and stiffness S of the specimen from the linear portion of the load-deflection plot.*
9. Conduct steps 1-8 on one low carbon steel and one cast iron specimen.
10. *Statistically compare the Young's moduli determined in bending tests and those in tensile tests.*

Procedure 4: 3 Point Bending Tests - bones

In this procedure, you will obtain the load-deflection data for a biological tissue: cortical bone. Segments of cortical chicken bone will be stripped of flesh and subjected to load-testing. These measurements will be used to determine the Young's modulus, which is similarly relevant for biological tissues as it is for metals. Finally, you will compare the materials properties of cooked to uncooked chicken bone.

For each member in you group, conduct the test with at least one raw drumsticks and one cooked drumsticks.

All students should exercise caution when using the scissors or scalpels and should wear protective goggles during this experiment.

1. Using scissors or other appropriate tool, remove the muscle and connective tissue from the chicken drumsticks.
2. Place a specimen on the 3-point bending device. The span depends on the length of the specimen. A span of 40 mm is recommended for most chicken leg bones. Measure and record the span.

3. Orient the specimen in such a manner that the maximum diameter (the most stable surface) is oriented as the base. Place a piece of paper towel under the specimen as the adipose on the surface of the bone and marrow within it will be somewhat messy.
4. Similar to Procedure 3, conduct the bending test until specimen fracture. *Record the fracture load and crosshead speed.*
5. Rename and save the data immediately after the test.
6. Remove the specimen from the test fixture. Completely break the bone and wash out any bone marrow in the canal.
7. Using a caliper, measure the inner and outer diameters of the bone. Repeat 3 times at varying circumferential positions and use the mean value as the actual dimension. Also record the circularity (variance from mean value) of the inner and outer surfaces. *Use these dimensions to determine the maximum strain and flexural ultimate stress of the bone specimen.*
8. Using your load-deflection data to calculate the Young's moduli for both uncooked and cooked chicken bones. *Determine if there is a statistically significant difference between the E of the raw vs. cooked bones. Account for any factors that might affect the measurements and explain any differences.*

Laboratory Report Guidelines

Be sure you include the following in your lab report:

- Answer/Discuss all of the italicized and underlined portions of the procedure and include all of the requisite plots to display your data (use the appendices if space becomes limited).
- Compare the Young's moduli of cast iron, low carbon steel, PE, PS, and bone, comment on the results of these comparisons with respect to your expectations.
- For tensile testing and 3-point bending testing, *provide sample calculations* to determine the requisite curves and properties from the load-displacement data.
- Ethics Question: You are charged with characterizing the mechanical properties of a femur for the purposes of designing a novel hip implant. What would you propose would be the optimal sample source(s) to best estimate the in situ properties of the bone. Factors to consider include (but are not limited to):
  - Source of the bone (human, animal, cultured)

- Condition of donor (live, cadaver, age of donor, pre-existing conditions).
- Manipulation of tissue (freshly explanted, fixed, etc.)
- Number of samples needed

In presenting your case, consider the likelihood and ethics of your ideal sample.

References/Relevant Reading

1. Chinese National Standard. Metallic materials – Tensile testing – Part 1: Method of test at room temperature (GB/T 228.1-2010). Standardization Administration of China. 2010.
2. Chinese National Standard. Plastics – Determination of tensile properties – Part 2: Test conditions for moulding and extrusion plastics (GB/T 1040.2-2006). Standardization Adminstration of China. 2006.
3. Willinghamm MD, Brodt MD, Lee KL, Stephens AL, Ye J, Silva MJ 2010 Age-related changes in bone structure and strength in female and male BALB/c mice. Calcif Tissue Int 86(6):470-483.